NIKHIL DEEKONDA

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EDUCATION

Yeshiva University, Katz School of Health and Science, NY, USA Sep 2023 - May 2025
Masters of Science in Artificial Intelligence - Sacks Imapct Scholar Fellowship **CGPA** 3.9

WORK EXPERIENCE

Valuai.io, Los Angeles, California, USA May 2025 – Present
AI Engineer

- Developed an AI-driven patient engagement platform that empowers clinicians to securely communicate with patients, analyze health data, and deliver personalized care recommendations-integrating AWS Bedrock, LangChain, Redis, and MongoDB for scalable, privacy-compliant deployment.
- Architected a Mixture of Experts (MoE) architecture on top of a DeepSeekR1-distilled LLaMA 8B model, introducing domain-specific experts fine-tuned with RAG-based datasets, improving contextual accuracy by approximately 30% in internal evaluations.
- Engineered a high-performance RAG pipeline using HNSW indexing for dense retrieval and cross-encoder reranking for contextual relevance, achieving 40% faster retrieval speed, improved factual grounding across clinical domains.
- Designed and currently expanding a Neo4j knowledge graph to enable structured reasoning and relationship-based querying, enhancing clinical recommendation precision and supporting longitudinal patient data insights.
- Fine-tuned large language models for the patient-side conversational interface using synthetic data generated via few-shot prompting with GPT-4o and DeepSeek, ensuring context-aware, guideline-aligned multi-turn interactions that reduced error rates by 20%.
- Built and deployed backend microservices using FastAPI, containerized with Docker, and orchestrated on AWS EC2-integrating AWS Cognito for secure authentication and Redis caching for low-latency session management, cutting API response time by 35%.

LTIMindtree, Pune, Maharashtra, India Jul 2022 - Jul 2023
Senior Software Engineer

- Engineered an enterprise-grade AI platform enabling organizations to safely deploy, monitor, and fine-tune large language models (LLMs) for real-world applications such as PII redaction, tone classification, and toxic content detection—driving automation accuracy and compliance improvements across departments.
- Developed robust guardrails, data pipelines, and validation workflows for ML/DL models including GPT, Falcon, LLaMA-2, and Mistral, leveraging LoRA and QLoRA fine-tuning to improve model performance and reduce resource costs by 35%.
- Designed and executed advanced Retrieval-Augmented Generation (RAG) pipelines connecting LLMs with vector databases, while creating hallucination detection modules using Small Language Models (SLMs) to lower factual error rates by 20%.
- Trained a proprietary 350M-parameter model from scratch and deployed fine-tuning, RAG, and validation workflows on AWS infrastructure (EC2, S3, SageMaker, Bedrock), achieving a 40% improvement in model training throughput and operational scalability.

National Highway Authority of India, Tirupati, Andhra Pradesh, India Jun 2021 - Aug 2021
Machine Learning Intern

- Developed predictive machine learning models to analyze national highway road conditions, enabling identification of key factors contributing to road deterioration, improving maintenance decision-making for infrastructure teams.
- Applied multiple regression and ensemble algorithms on a dataset of over 500,000 road instances, achieving strong model performance (high R-squared values) and accurately forecasting rutting severity across diverse terrains.
- Designed and optimized end-to-end preprocessing and feature engineering pipelines using Python and Scikit-learn, reducing prediction error for rutting analysis by more than 15% accelerating analytical turnaround time by 40%.
- Delivered actionable insights to the National Highway Authority of India through feature impact analysis, supporting data-driven planning and prioritization of road maintenance projects.

PROJECTS

Mixture-of-Experts (MoE) Implementation in PyTorch

- Designed and developed scalable MoE architectures in PyTorch to enhance the efficiency and performance of LLM research and experimentation across both single-device and distributed multi-GPU environments.
- Built and optimized three distinct MoE variants—Basic, Sparse, and Shared Expert Sparse—demonstrating advanced proficiency in expert routing, specialization, and large-scale model parallelization.
- Implemented token-level dynamic gating using top-k selection and tensor-based dispatching to intelligently route inputs to the most relevant experts, improving compute utilization and reducing redundant computation.
- Introduced a hybrid load-balancing loss function combining auxiliary loss and z-loss, resulting in more uniform expert utilization and a 25% reduction in training instability across runs.
- Deployed distributed training using PyTorch `DistributedDataParallel`, achieving a 40% improvement in overall training throughput and maintaining expert synchronization across GPUs and nodes.
- Leveraged PyTorch autograd to implement custom backward passes within the gating mechanism, reducing gradient memory overhead by 15% and improving overall model training efficiency.

TaskFin: RAG-Based Financial Task Automation

- Designed and implemented a Retrieval-Augmented Generation system for end-to-end bill payment automation, enabling users to complete financial tasks through natural language interactions with high contextual accuracy.
- Developed a modular multi-agent architecture featuring a central Orchestrator Agent that coordinated specialized sub-agents for Authentication, Financial Transactions, and State Management—leveraging LangChain memory types (buffer, entity, summary) and MCP servers to preserve contextual state across conversations.
- Authored comprehensive Product Requirement Documents (PRDs) to establish measurable objectives, streamline development cycles, and ensure alignment between technical implementation and user experience goals.
- Deployed the complete solution as a fully interactive Streamlit conversational application, enabling secure data handling, real-time RAG-driven task automation, and rapid prototyping of AI-powered fintech assistants.
- Optimized agent orchestration and retrieval latency, achieving a 30% improvement in response time and a 25% reduction in context drift during multi-turn financial interactions.

LungAware: AI Lung Cancer Detection App

- Engineered a deep convolutional neural network (CNN) for early lung cancer detection, achieving 99.4% classification accuracy through advanced image preprocessing, transfer learning, and hyperparameter optimization.
- Developed cross-platform (Android/iOS) mobile applications with Swift (iOS) and Kotlin (Android), integrating TFLite-quantized models and Grad-CAM for explainable AI and transparent diagnostic support.
- Implemented a seamless user interface and cloud-based backend (AWS EC2) for automated inference, patient management, and result notification, ensuring compliance with healthcare privacy and security best practices.

SKILLS

Programming Languages	Python, Swift, Java, JavaScript, R, C, C++
Frameworks / Libraries	PyTorch, TensorFlow, FastAPI, LangChain, CrewAI, LangGraph, CoreML
Databases	PostgreSQL, MySQL, MongoDB, Redis
Cloud / DevOps	AWS, GCP, Docker, Kubernetes, Git, GitHub
Tools / Analytics	Excel, Tableau, Jupyter, VS Code, XCode
Other	LLM Fine-tuning, RAG Pipelines, Multi-agent Systems, iOS Development (SwiftUI)

CERTIFICATIONS

- AWS Certified Machine Learning Engineer – Associate
- AWS Certified AI Practitioner
- Engineer Data in Google Cloud
- Exploring Data With Looker
- SQL-HackerRank
- Problem Solving-HackerRank